

COTIVITI GENAI INTERN ASSESSMENT · NOOPUR SHEKHAR DIVEKAR · JUNE 2026

Agentic Policy Intelligence for Prior Authorization

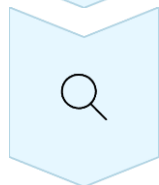
ClearAuth proof of concept — AI first pass, human final call

Prior Auth Is Slow, Manual, and Inconsistent

Reviewers hunt through lengthy policy PDFs for every case — creating delays, burden, and inconsistent decisions.



Request In



Manual PDF Hunt



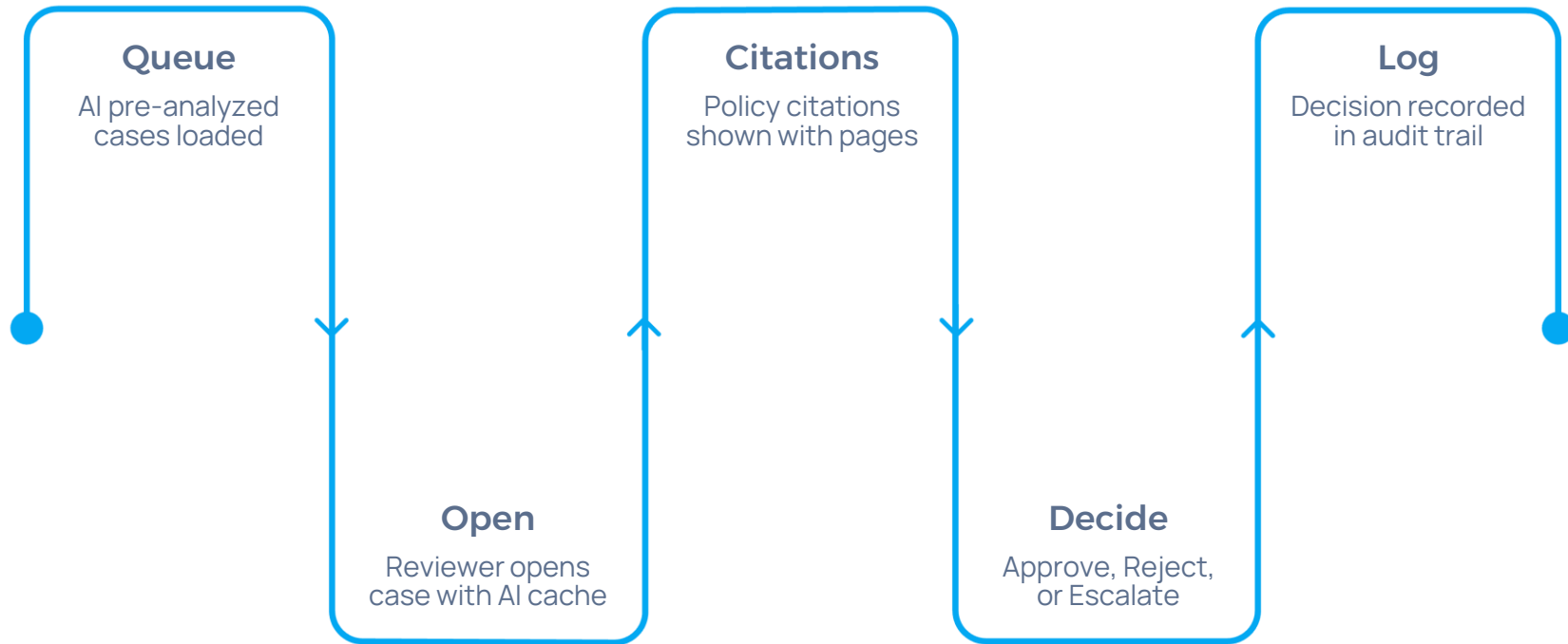
Decision Delay



AMA Survey: Prior auth is among the top administrative burdens for physicians and health plan staff.

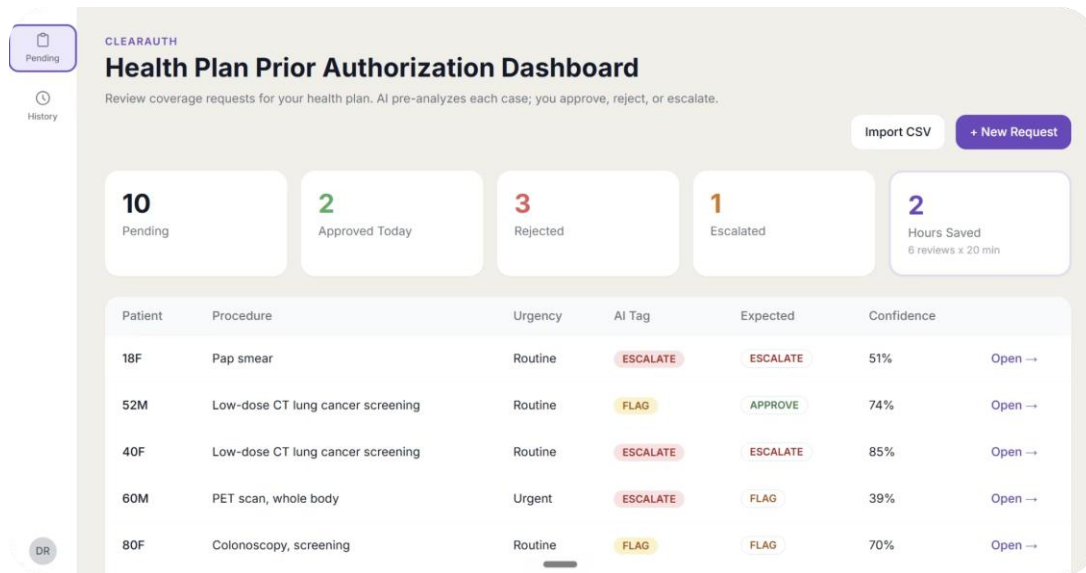
ClearAuth: AI First Pass, Human Final Call

Not auto-approval. The AI pre-analyzes every case before the reviewer opens it — cutting research time, not replacing judgment.



Every case is pre-analyzed before the reviewer opens it, so research time drops, human judgment stays central.

Live Dashboard: Instant Context on Open



Health Plan Prior Authorization Dashboard

Pending table shows AI Tag, Expected label, and Confidence score per case.

📄 Synthetic cases pre-analyzed at instant open, zero wait.

Case Detail: Evidence Before Action

Reviewer Sees Everything Before Deciding

[← Back to pending](#)

RejectEscalate✓ Approve

Low-dose CT lung cancer screening | 40F

ICD: Z12.2 · Lung cancer screening · Submitted just now

AI Recommendation
ESCALATE

Confidence
85%

Urgency
Routine

Analysis Time
19.7s

Clinical Request

Patient: 40F Urgency: Routine
ICD-10: Z12.2 CPT: 71271
Diagnosis: Lung cancer screening
Clinical Notes: 40F, never smoker, no risk factors

Review Chain

A

 AI Agent

Y

 You (Reviewer)

ESCALATE
Awaiting decision

Evidence & Rationale

Per Ncd103C1Part4 page 29, beneficiaries must be age 50–77 years. This patient is 40 years old, which falls below the minimum eligible age of 50. This criterion is NOT met. Per Ncd103C1Part4 page 29, beneficiaries must have a tobacco smoking history of at least 20 pack-years and be a current smoker or have quit within the last 15 years. The clinical notes document this patient as a 'never smoker.' This criterion is NOT met. Clinical notes state 'no risk factors.' The policy (Ncd103C1Part4 page 29) does not provide an alternative pathway

Source Documents

PDF

 Ncd103C1Part4
ncd103c1part4.pdf

Open PDFView excerpt (p.29)

View excerpt (p.29)View excerpt (p.29)

AI Recommendation + Confidence

APPROVE / FLAG / ESCALATE with cited policy excerpts and page numbers

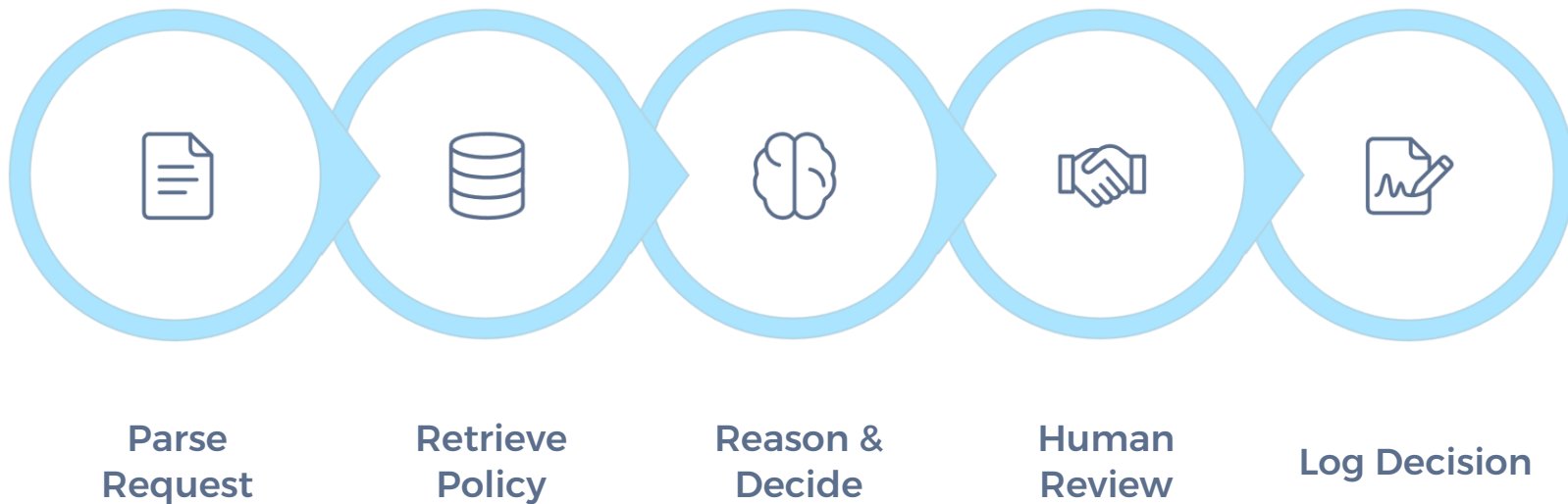
Clinical Request + Evidence

Full request context, rationale, and retrieved policy chunks side by side

Source Documents

Open PDF or jump to cited excerpt in one click

Architecture: LangGraph Agent + RAG Pipeline



Side components: ChromaDB (~1,749 CMS/NCD policy chunks), Claude API (claude-sonnet-4-6), SQLite audit trail. RAG pipeline: PDF → chunk/embed → vector search → cite excerpts.

Three Design Choices That Matter



RAG Grounding

Model reasons *only* over retrieved CMS/NCD policy excerpts — no free-form generation.



Anti-Hallucination Guards

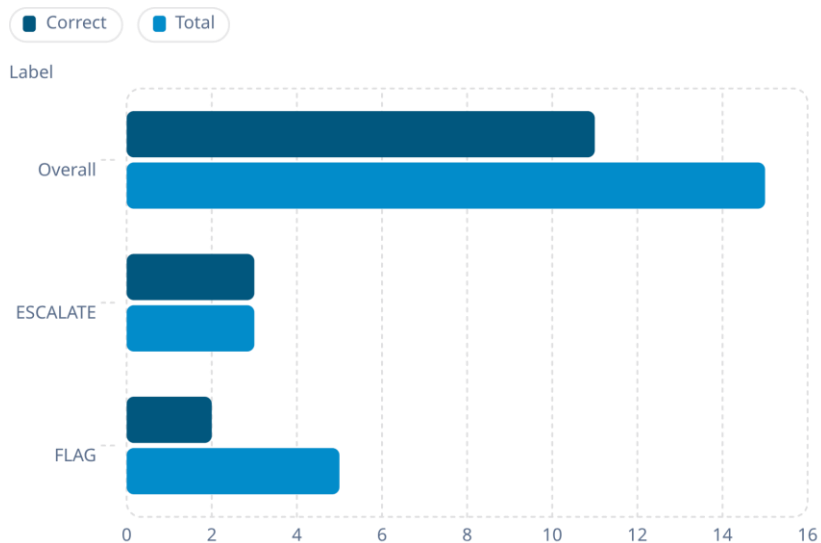
Undetectable or confidence < 60% → auto-escalate. No confident guesses.



Governance Built In

Synthetic data only. Audit tables. Human signature required on every decision.

Benchmark: Honest Results on Synthetic Data



73% Overall Match (11/15)

ESCALATE: 3/3 — perfect on clear-cut cases requiring escalation.

FLAG: 2/5 — borderline cases deferred to human review, by design.

Human reviewer always has final authority. AI recommends; humans decide.

Why This Matters for Cotiviti

Payment Accuracy & TPO Operations

Policy intelligence layer fits existing workflows – reduce manual review load immediately.

Reusable Architecture

Same RAG + agent pattern extends to coding audit, coverage determination, and beyond.

Pilot Recommendation

One plan, one policy set. Measure override rate and time saved – then scale.

AI Recommends with Evidence. Humans Decide with Accountability.

Live Demo

<http://localhost:8000>

FastAPI · LangGraph · ChromaDB · Claude · SQLite · Tailwind UI

50 synthetic cases · ~1,749 policy chunks

References

- AMA Prior Authorization Survey (2023)
- CMS National Coverage Determinations
- Lewis et al. Retrieval-Augmented Generation (2020)
- LangGraph Documentation (2025)
- ClearAuth POC — Noopur Shekhar (2026)